Devadarsh A Nair

B.E. (Hons.) Electronics and Electrical Engineering
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Education

BITS Pilani, K.K. Birla Goa Campus — *Goa, India*
B.E. (Hons.) Electronics and Electrical Engineering — 2024 – 2028
CGPA: 8.63 / 10

DEENS Academy — *Bangalore*
Class XII, CBSE — 2023
Percentage: 92.8%

Delhi Public School, East — *Bangalore*
Class X, CBSE — 2021
Percentage: 94.16%

Relevant Coursework

Microelectronic Circuits, Electronic Devices, Electronic Device Simulation Laboratory, Physics of Semiconductor Devices, Digital Design, Microprocessors & Interfacing, Signals & Systems, Control Systems, Electromagnetic Theory, Quantum Mechanics-I.

Technical Skills

Hardware Description Languages	Verilog
EDA / Simulation Tools	Sentaurus TCAD, Cadence Virtuoso, LTspice, Proteus, Simulink
Programming	MATLAB, Python, C, C++, x86 Assembly (MASM)
Platforms & Tools	Git, Linux, Verilator, Icarus Verilog, GTKWave, LaTeX

Projects

Digital Stopwatch Controller (Verilog + Verilator HW/SW Co-Design) — BITSilicon, Sem-4 2025–26
- Designed a synchronous digital stopwatch RTL in Verilog with a modular hierarchy: top module, seconds counter (0–59), minutes counter (0–99), and a control FSM (IDLE / RUNNING / PAUSED) supporting start, stop, resume, and reset.
- Enforced fully synchronous, synthesizable RTL — non-blocking assignments, synchronous up-counters (no ripple), and active-low reset — verified with a self-checking Verilog testbench.
- Built a cycle-accurate C++ control program using Verilator to drive the DUT; validated reset, pause/resume, and periodic MM:SS readout through the top-level interface.

Synchronous FIFO with Self-Checking Verification Environment — BITSilicon, Sem-4 2025–26
- Implemented a parameterized synchronous FIFO (DATA_WIDTH, DEPTH) in Verilog with write/read pointers, an occupancy counter, and correct `wr_full`/`rd_empty` flag logic including simultaneous read-write handling.
- Built a self-checking testbench with an independent golden reference model and a scoreboard performing per-cycle comparison of `rd_data`, `count`, and flag signals, with detailed failure reporting on mismatch.
- Wrote directed tests (reset, fill, drain, overflow, underflow, simultaneous RW, pointer wrap-around) and manual coverage counters to confirm all critical corner cases were exercised.

Rectifying Circuits: Schottky vs. PN Junction Diode Comparison — Sem-4, 2025–26
- Characterized forward voltage drop, reverse recovery time, and rectification efficiency across half-wave and full-wave topologies using LTspice and bench measurements.
- Quantified trade-offs for low-voltage and high-frequency applications, highlighting Schottky advantages in switching regimes versus PN robustness under reverse bias.

Adaptive Noise Cancellation using MATLAB — Sem-4, 2025–26
- Implemented an LMS-based adaptive filter for real-time suppression of additive noise from a reference-corrupted signal using MATLAB and the DSP System Toolbox.
- Studied convergence behavior and steady-state MSE across varying step sizes and filter orders; validated SNR improvement on synthetic and audio test signals.

Landau Levels and the Integer Quantum Hall Effect — Sem-4, 2025–26
- Derived quantization of energy levels for a 2D electron gas in a perpendicular magnetic field and analyzed the resulting Hall conductance plateaus.
- Connected the theoretical framework to semiconductor device behavior in high-mobility 2DEG systems and metrological resistance standards.

8086-Based Whack-a-Mole Game — Sem-4, 2025–26
- Designed and implemented an interactive game on the 8086 microprocessor with assembly-level interfacing via 8255 PPI, timers, and interrupts (MASM, Proteus).
- Integrated LED output and push-button input with scoring logic and randomized event generation using the 8253 programmable timer.

Clubs & Activities

- **BITSilicon Semiconductor Club, BITS Pilani Goa** — Member. Engaged in semiconductor device simulation, analog/mixed-signal design sessions, and club-led RTL/verification assignments.
- **TEDxBITSGoa** — Core Member. Contributed to event organization, curation, and production for the campus TEDx chapter.

Achievements

- **NTSE Scholar, 2021** — Awarded by NCERT, Government of India; one of ˜2,000 scholars selected nationally across lakhs of candidates.

Extracurriculars & Interests

Freelance graphic designer and video editor — commissioned work across social, event, and promotional content. Interests: analog & mixed-signal IC design, semiconductor device physics & modelling, condensed matter physics, quantum physics.